Cumulative frequency tables

1 Consider the following table:

a Find the total number of scores recorded.

b Find the number of times a score of 14 occurred.

c Find the number of times a score less than 13 occurred.

Score (x)	Cumulative frequency (cf)
10	7
11	15
12	18
13	20
14	26

2 Complete the given frequency table:

Score (x)	Frequency (f)	Cumulative frequency (cf)
12	8	8
13	5	
14	8	21
15		26

3 For the given frequency table:

a Complete the cumulative frequency column.

b Calculate the total frequency.

c State the class size.

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1 \rightarrow 4$	4	
$5 \rightarrow 8$	5	
$9 \rightarrow 12$	9	
$13 \rightarrow 16$	5	
$17 \rightarrow 20$	4	

4 For the given frequency table:

a Complete the cumulative frequency column.

b Calculate the total frequency.

c State the class size.

d Approximately half of the scores recorded are greater than what score?

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1 \rightarrow 5$	15	
$6 \rightarrow 10$	26	
$11 \rightarrow 15$	18	
$16 \rightarrow 20$	14	
$21 \rightarrow 25$	7	
$26 \rightarrow 30$	2	

5 For the given frequency table:

a Complete the cumulative frequency column.

b Calculate the total frequency.

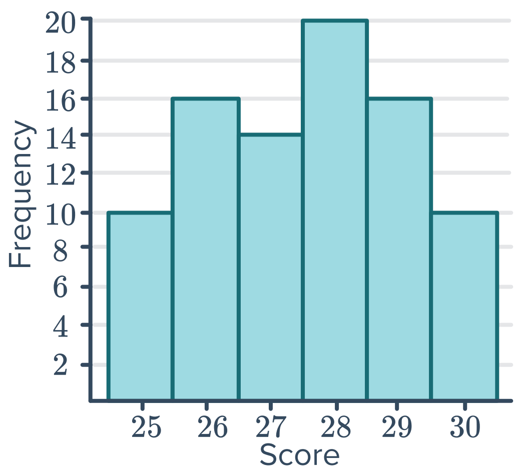
c State the class size.

d Approximately one third of the scores recorded are greater than what score?

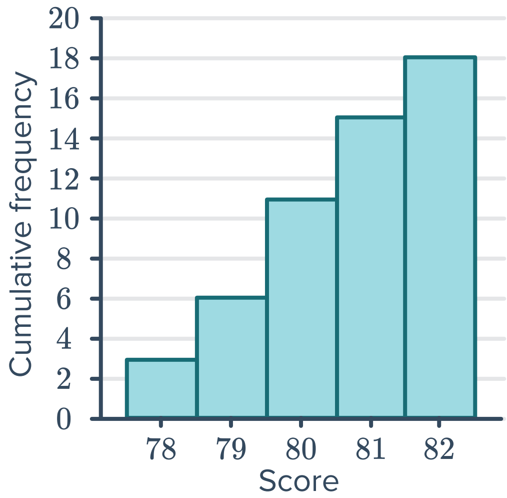
Score (x)	Frequency (f)	Cumulative frequency (cf)
$20 \rightarrow 24$	7	
$25 \rightarrow 29$	18	
$30 \rightarrow 34$	25	
$35 \rightarrow 39$	12	
$40 \rightarrow 44$	8	
$45 \rightarrow 49$	4	
$50 \rightarrow 54$	1	

Cumulative frequency graphs

6 Construct a cumulative frequency table for the data represented in the given histogram:



7 Construct a frequency table for the data represented in the given cumulative frequency histogram:



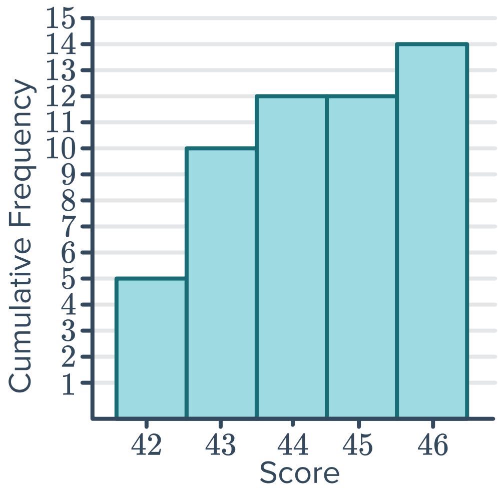
8 Consider the following cumulative frequency histogram:

a Find the total number of scores recorded.

b Find the number of times a score of 46 occurred.

c Find the number of times a score of 45 occurred.

d Find the percentage of scores that were 43 or less. Round your answer to one decimal place.



9 Consider the frequency table showing the number of 'holes in one' across golf tournaments:

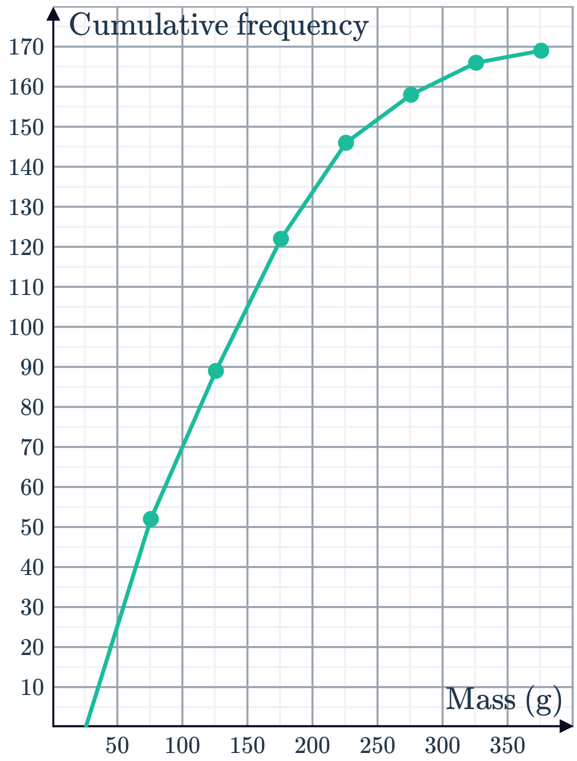
a Construct a cumulative frequency histogram for this data.

b Find the total number of 'holes in one' across all the tournaments.

c In how many tournaments were at most 3 'holes in one' scored?

Number of holes in one	Tournaments
2	5
3	1
4	3
5	4
6	0

10 The following graph shows the cumulative frequency ogive of the masses of fish caught in a fishing competition:
Construct a frequency histogram to depict the distribution of the masses of the fish.



11 The marks received by a group of 20 students are given below:

$$55, 90, 90, 99, 78, 78, 78, 99, 78, 99, 55, 99, 78, 64, 64, 78, 78, 99, 64, 78$$

a Construct a cumulative frequency histogram to represent the data.

b How many students scored above 70?

c How many students scored less than or equal to 70?

d How many students scored above 80 but less than or equal to 90?

Applications

12 A principal wants to investigate the performance of students at his school in Performing Arts. To do this, he has the marks of each student studying Performing Arts collected into groups and put into a frequency table. Each group of marks is assigned a grade as shown in the following frequency table:

Grade	Score (x)	Frequency (f)	Cumulative frequency (cf)
E	$0 \leq x < 20$	7	
D	$20 \leq x < 40$	14	
C	$40 \leq x < 60$	32	
B	$60 \leq x < 80$	97	
A	$80 \leq x < 100$	62	

a Complete the table by finding the cumulative frequency values.

b Calculate the total frequency.

c State the class size.

d Approximately three quarters of the scores recorded are greater than what score?

13 A pair of dice are rolled 50 times and the numbers appearing on the uppermost face are added to give a score. The results are recorded in the given table:

Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	
3	2	
4	5	
5	5	
6	5	
7	9	
8	7	
9	5	
10	8	
11	1	
12	2	

a State the lowest possible score when a single pair of dice are rolled.

b State the highest possible score when a single pair of dice are rolled.

c Complete the table by finding the cumulative frequency values.

d Find the number of times a score of 8 occurred.

e Find the number of times a score more than 9 occurred.

f Find the number of times a score of at most 6 occurred.

14 The number of sightings of the Northern L[redacted] were recorded across various Canadian locations over a period of 1 month. The list below represents the number of sightings at each location:

$$11, 10, 10, 9, 7, 8, 8, 12, 12, 12, 12, 12, 12, 9, 9, 12, 9, 9, 8, 8$$

a Complete the table.

b In how many locations were there at least 8 sightings?

c In how many locations were there less than 11 sightings?

Number of sightings	Number of locations (f)	Cumulative frequency (cf)
7		
8		
9		
10		
11		
12		

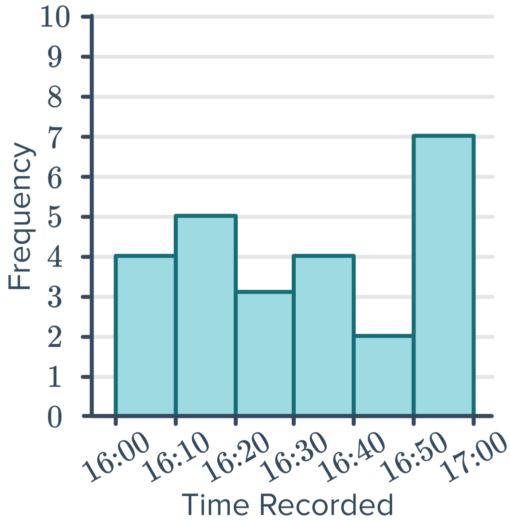
15 A 1500 m swimmer records her time over several training sessions. Her times are recorded in the following histogram:

a Construct a cumulative frequency table for the data using the given intervals.

b Find the total number of training sessions she completed.

c Find the number of times she recorded a swim time faster than 16:40.

d Find the percentage of swims that were less than 16:30.



16 The heights of 22 boys in a class are listed:

$$164, 167, 158, 159, 150, 166, 150, 146, 149, 161, 164,$$
$$163, 152, 161, 157, 157, 153, 157, 156, 165, 162, 161$$

a Construct a cumulative frequency histogram for the data. Use the discrete intervals of $146 - 150$, $151 - 155$, etc.

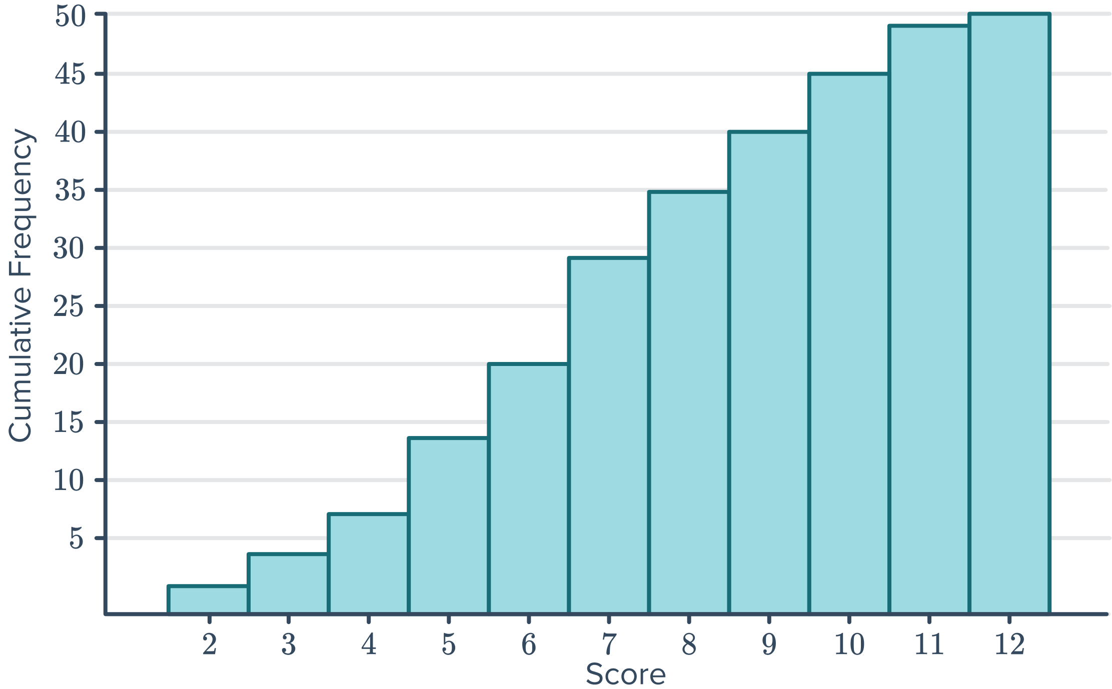
b How many students are taller than 155 cm?

c How many students are at most 155 cm tall?

d How many students are taller than 150 cm but shorter than 156 cm?

e Calculate the centre of the class $151 - 155$.

17 A pair of dice are rolled 50 times and the [redacted]ers appearing on the uppermost face are added to give a score. The results are presented in the following histogram:



a Explain how the frequency of a certain score can be determined from a cumulative frequency histogram.

b State the most frequent score.